



Yakeen Plus (2025)

Practice Test - 10

DURATION : 200 Minutes

DATE : 24/11/2024

M. MARKS : 720

Topics Covered

Physics :	Physics and Measurement, Kinematics, Laws of Motion, Work, Energy, and Power, Rotational Motion
Chemistry :	Some Basic Concept of Chemistry, State of Matter, Solutions, Organic Chemistry: Some Basic principles and Techniques, (IUPAC Naming), Organic Chemistry: Some Basic, principles and Techniques, (Isomerism), Organic Chemistry: Some Basic principles and Techniques, (GOC), Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Biology :	(Botany): Cell - The Unit of Life, Cell Cycle and Cell Division, The Living World, Biological Classification, Plant Kingdom (Zoology): Structural Organization in Animals, Animal Kingdom, Biomolecules, Breathing and Exchange of Gases

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

SECTION - A

1. A wheel of radius 1 meter rolls forward half a revolution on a horizontal ground. The magnitude of the displacement of the point of the wheel initially in contact with the ground is:

(1) 2π (2) $\sqrt{2}\pi$
 (3) $\sqrt{\pi^2 + 4}$ (4) π

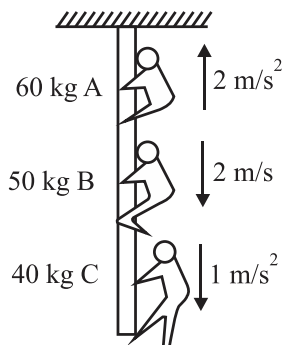
2. If a cycle wheel of radius 4 m completes one revolution in two seconds. Then acceleration of a point on the cycle wheel will be:

(1) $\pi^2 m/s^2$ (2) $2\pi^2 m/s^2$
 (3) $4\pi^2 m/s^2$ (4) $8\pi m/s^2$

3. What is the value of linear velocity, if $\vec{\omega} = 3\hat{i} - 4\hat{j} + \hat{k}$ and $\vec{r} = 5\hat{i} - 6\hat{j} + 6\hat{k}$

(1) $6\hat{i} + 2\hat{j} - 3\hat{k}$ (2) $-18\hat{i} - 13\hat{j} + 2\hat{k}$
 (3) $4\hat{i} - 13\hat{j} + 6\hat{k}$ (4) $6\hat{i} - 2\hat{j} + 8\hat{k}$

4. Tension in the rope at the rigid support is: ($g = 10 \text{ m/s}^2$)



(1) 760 N (2) 1360 N
 (3) 1580 N (4) 1620 N

5. Match List-I with List-II:

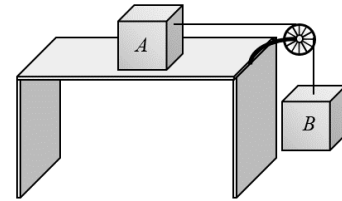
A man of mass m is on the floor of a lift then match the following.

List-I		List-II	
(A)	Lift is moving up with acceleration a	(I)	Apparent weight is greater than true weight
(B)	Lift is moving down with acceleration a	(II)	Apparent weight is zero
(C)	Lift is moving with uniform velocity	(III)	Apparent weight is equal to true weight
(D)	Lift is freely falling	(IV)	Apparent weight is less than true weight

Choose the **correct** answer from the options given below:

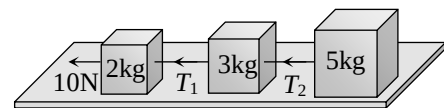
(1) A-II, B-III, C-IV, D-I
 (2) A-I, B-IV, C-III, D-II
 (3) A-II, B-IV, C-III, D-I
 (4) A-III, B-II, C-IV, D-I

6. A block A of mass 7 kg is placed on a frictionless table. A thread tied to it passes over a frictionless pulley and carries a body B of mass 3 kg at the other end. The acceleration of the system is: (given $g = 10 \text{ ms}^{-1}$)



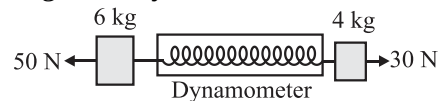
(1) 100 ms^{-2} (2) 3 ms^{-2}
 (3) 10 ms^{-2} (4) 30 ms^{-2}

7. Three blocks of masses 2 kg, 3 kg and 5 kg are connected to each other with light string and are then placed on a frictionless surface as shown in the figure. The system is pulled by a force $F = 10 \text{ N}$, then tension $T_1 =$



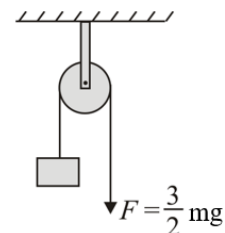
(1) 1 N (2) 5 N
 (3) 8 N (4) 10 N

8. A dynamometer D is attached to two blocks of masses 6 kg and 4 kg as shown in the figure. The reading to the dynamometer is:



(1) 18 N (2) 28 N
 (3) 38 N (4) 48 N

9. In the arrangement shown, the mass m will ascend with an acceleration: (Pulley and rope are massless)

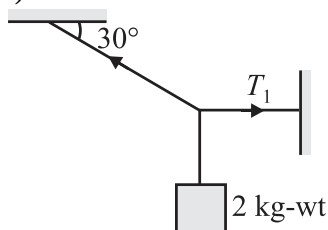


(1) Zero (2) $g/2$
 (3) g (4) $2g$

10. Given below are two statements:
Statement-I: Linear momentum of a body changes even when it is moving uniformly in a circle.
Statement-II: Force required to move a body uniformly along a straight line is zero.
 In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

11. A body of weight 2 kg is suspended as shown in the figure. The tension T_1 in the horizontal string (in kg wt) is:



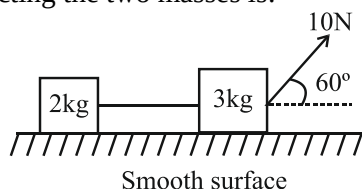
- (1) $\frac{2}{\sqrt{3}}$
- (2) $\frac{\sqrt{3}}{2}$
- (3) $2\sqrt{3}$
- (4) 2

12. The tension in the spring is:



- (1) Zero
- (2) 2.5 N
- (3) 5 N
- (4) 10 N

13. Figure shows two blocks connected by a light inextensible string as shown in figure. A force of 10 N is applied on the bigger block at 60° with horizontal, then the tension in the string connecting the two masses is:



- (1) 5 N
- (2) 2 N
- (3) 1 N
- (4) 3 N

14. A particle is projected with speed u at an angle θ with the horizontal. Find the radius of curvature at the highest point of its trajectory.

- (1) $\frac{u^2 \cos^2 \theta}{2g}$
- (2) $\frac{\sqrt{3}u^2 \cos^2 \theta}{2g}$
- (3) $\frac{u^2 \cos^2 \theta}{g}$
- (4) $\frac{\sqrt{3}u^2 \cos^2 \theta}{g}$

15. The velocity of a body depends on time according to the equation $v = 20 + 0.1 t^2$. The body is undergoing:

- (1) uniform acceleration.
- (2) uniform retardation.
- (3) non-uniform acceleration.
- (4) zero acceleration.

16. A projectile fired with initial velocity u at some angle θ has a range R . If the initial velocity be doubled at the same angle of projection, then the range will be:

- (1) $2R$
- (2) $R/2$
- (3) R
- (4) $4R$

17. The equation of projectile is $y = 16x - \frac{5x^2}{4}$. The horizontal range is:

- (1) 16 m
- (2) 8 m
- (3) 3.2 m
- (4) 12.8 m

18. Two cars of masses m_1 and m_2 are moving along the circular paths of radius r_1 and r_2 respectively. The speed are such that they complete one round at the same time. The ratio of angular speeds of two cars is:

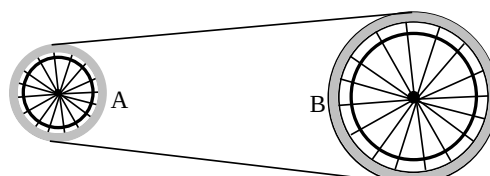
- (1) $m_1 : m_2$
- (2) $r_1 : r_2$
- (3) 1 : 1
- (4) $m_1 r_1 : m_2 r_2$

19. A particle moving along the circular path with a speed v at time t_0 and its speed increases by g in one second. If the radius of the circular path be r , then the net acceleration of the particle at t_0 is:

- (1) $\frac{v^2}{r} + g$
- (2) $\frac{v^2}{r^2} + g^2$
- (3) $\left[\frac{v^4}{r^2} + g^2 \right]^{1/2}$
- (4) $\left[\frac{v^2}{r} + g \right]^{1/2}$

20. A continuous conveyor belt passes over two discs A and B of radii R and $4R$ respectively. They start from rest. A has angular acceleration of $\frac{\pi}{4} \text{ rad s}^{-2}$.

What is the angular velocity of the disc B at the end of 32 sec?

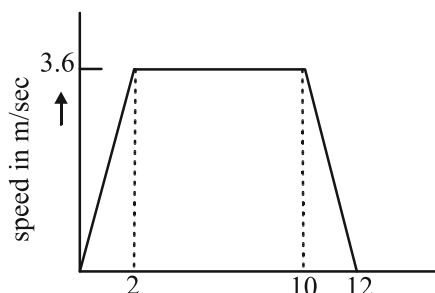


- (1) $\pi \text{ rad s}^{-1}$
- (2) $2\pi \text{ rad s}^{-1}$
- (3) $4\pi \text{ rad s}^{-1}$
- (4) $\frac{\pi}{4} \text{ rad s}^{-1}$

21. A cubical block rests on a plane of $\mu = \sqrt{3}$. The angle through which the plane be inclined to the horizontal so that block just slides down will be:

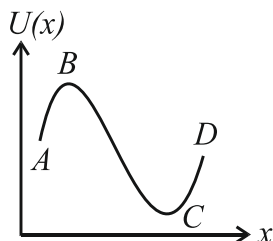
(1) 30° (2) 45°
(3) 60° (4) 75°

22. A lift is going up. The total mass of the lift and the passenger is 1500 kg. The variation in the speed of the lift is as given in the graph. The tension in the rope pulling the lift at $t = 11^{\text{th}}$ sec will be:



(1) 17400 N
(2) 14700 N
(3) 12000 N
(4) Zero

23. The potential energy of a particle varies with distance x as shown in the graph. The force acting on the particle is zero at.



(1) C
(2) B
(3) B and C
(4) A and D

24. A body of mass 6 kg under a force which caused displacement in it given $S = \frac{t^2}{4}$, metres where 't' is time. The work done by the force in 2 seconds is:

(1) 12 J (2) 9 J
(3) 6 J (4) 3 J

25. The potential energy U varies with the position x of a particle according to relation $U(x) = (x^2 - 6x)$ J (where x is in meter). Choose the **correct** option.

(1) $x = 3$ is a point of unstable equilibrium.
(2) $x = -2$ is a point of stable equilibrium.
(3) $x = 2$ is a point of equilibrium.
(4) $x = 3$ is a point of stable equilibrium.

26. A physical quantity P is given by $P = \frac{A^3 B^{\frac{1}{2}}}{C^{-4} D^{\frac{3}{2}}}$.

The quantity which brings maximum percentage error in P is:

(1) A (2) B
(3) C (4) D

27. A quantity is represented by $X = M^a L^b T^c$. The % error in measurement of M , L and T are $\alpha\%$, $\beta\%$ and $\gamma\%$ respectively. The maximum % error in X would be:

(1) $(\alpha a + \beta b + \gamma c)\%$
(2) $(\alpha a - \beta b + \gamma c)\%$
(3) $(\alpha a - \beta b - \gamma c) \times 100\%$
(4) None of these

28. The dimensions of $\frac{\alpha}{\beta}$ in the equation $F = \frac{\alpha - t^2}{\beta v^2}$,

where F is the force, v is velocity and t is time, is

(1) $[MLT^{-1}]$ (2) $[ML^{-1}T^{-2}]$
(3) $[ML^3T^{-4}]$ (4) $[ML^2T^{-4}]$

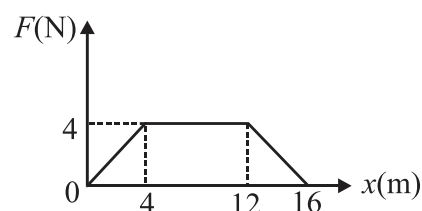
29. A force F is given by $F = at + bt^2$, where t is time. What are the dimensions of a and b

(1) MLT^{-3} and ML^2T^{-4}
(2) MLT^{-3} and MLT^{-4}
(3) MLT^{-1} and MLT^0
(4) MLT^{-4} and MLT^1

30. In a new system of units, unit of mass is x kg, unit of length is y metre and unit of time is z second. Now if 1 newton = F new units then $F =$

(1) $\frac{z}{xy}$ (2) $\frac{z^2}{xy}$
(3) $\frac{z}{xy^2}$ (4) $\frac{z}{x^2y}$

31. The figure given below shows the force F versus displacement x graph. The net work done by the force when an object is displaced from $x = 0$ to $x = 16$ m is

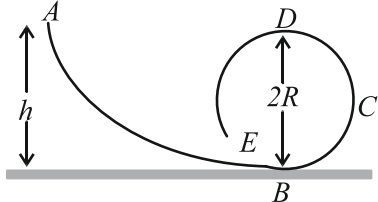


(1) 84 J (2) 24 J
(3) 48 J (4) 100 J

32. If the linear momentum is increased by 50%, the kinetic energy will increase by:

(1) 50% (2) 100%
(3) 125% (4) 25%

33. A frictionless track $ABCDE$ ends in a circular loop of radius R . A body slides down the track from point A which is at a height $h = 5$ cm. Maximum value of R for the body to successfully complete the loop is:



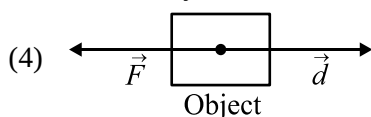
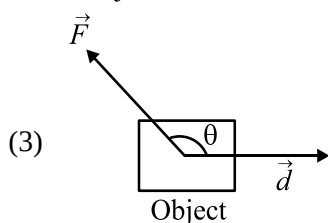
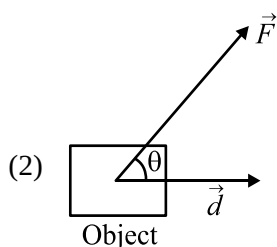
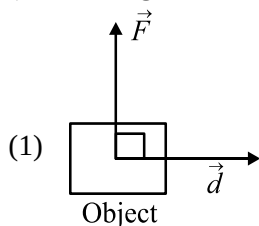
(1) 5 cm (2) $\frac{15}{4}$ cm
(3) $\frac{10}{3}$ cm (4) 2 cm

34. The potential energy of a body is given by, $U = A - Bx^2$ (where x is the displacement). The magnitude of force acting on the particle is

(1) Constant
(2) Proportional to x
(3) Proportional to x^2
(4) Inversely proportional to x

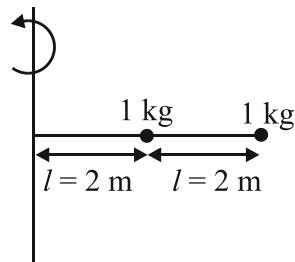
35. In which of the following figures the work done by force $\vec{F}$ in displacing an object through distance d is positive?

(θ is the angle between $\vec{F}$ and $\vec{d}$)



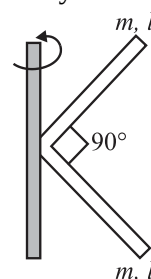
SECTION - B

36. Moment of inertia of point masses about given axis as shown will be:



(1) 8 kg.m^2 (2) 16 kg.m^2
(3) 20 kg.m^2 (4) 4 kg.m^2

37. Two identical uniform rods each of mass m and length l are joined perpendicular to each other. An axis passes through junction and in the plane of rods. Then M.I. of system about the axis is:

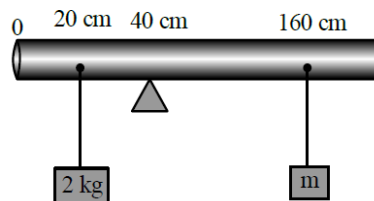


(1) $\frac{1}{3} ml^2$ (2) $\frac{1}{3\sqrt{2}} ml^2$
(3) ml^2 (4) $\frac{ml^2}{2}$

38. A thin rod of mass M and length L is bent into a circular ring. The expression for moment of inertia of ring about an axis passing through its diameter is:

(1) $\frac{ML^2}{2\pi^2}$ (2) $\frac{ML^2}{4\pi^2}$
(3) $\frac{ML^2}{8\pi^2}$ (4) $\frac{ML^2}{\pi^2}$

39. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass ' m ' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of ' m ' such that the rod is in equilibrium: ($g = 10 \text{ m/s}^2$)



(1) $\frac{1}{2} \text{ kg}$ (2) $\frac{1}{3} \text{ kg}$
(3) $\frac{1}{6} \text{ kg}$ (4) $\frac{1}{12} \text{ kg}$

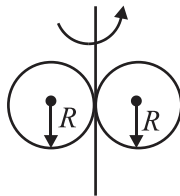
40. Two cylinders, one of which is hollow and other solid, have same mass and same moment of inertia about their respective geometrical axis. The ratio of their radii is:

- (1) $1:\sqrt{2}$ (2) $1:\sqrt{3}$
(3) $\sqrt{3}:\sqrt{3}$ (4) $1:2$

41. A body at rest breaks into two pieces of equal masses. The parts will move:

- (1) in same direction
(2) along different lines
(3) in opposite directions with equal speed
(4) in opposite directions with unequal speeds

42. Moment of inertia of combination of two discs of same mass M and same radius R kept in contact about the tangent passing through point of contact and in the plane of discs, as shown is:

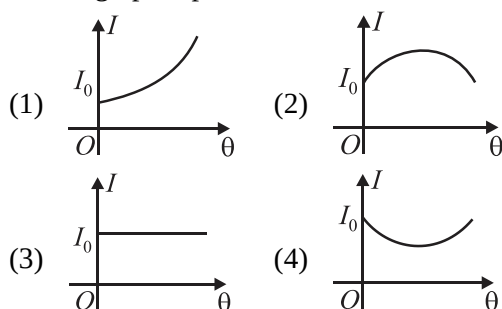


- (1) $\frac{MR^2}{4}$ (2) $\frac{5}{4}MR^2$
(3) $\frac{MR^2}{2}$ (4) $\frac{5}{2}MR^2$

43. The moment of inertia of a uniform semicircular wire of mass m and radius r , about an axis passing through its centre of mass and perpendicular to its plane is:

- (1) $\frac{mr^2}{2}$ (2) mr^2
(3) $mr^2\left(1 - \frac{4}{\pi^2}\right)$ (4) $mr^2\left(1 + \frac{4}{\pi^2}\right)$

44. A square plate has a moment of inertia I_0 about an axis lying in its plane, passing through its centre and making an angle θ with one of the sides. Which graph represents the variation of I with θ ?



45. A person of mass m is standing on one end of a plank of mass M and length L and floating in water. The person moves from one end to another and stops. The displacement of the plank is:

- (1) $\frac{Lm}{(m+M)}$
(2) $Lm(M+m)$
(3) $L\frac{(m+M)}{m}$
(4) $\frac{LM}{(m+M)}$

46. The angular velocity ω of a particle varies with time t as $\omega = (10t^2 + 15)\text{rad/s}$. The angular acceleration of the particle at $t = 1\text{s}$ is:

- (1) 10 rad/s^2 (2) 20 rad/s^2
(3) 5 rad/s^2 (4) Zero

47. The angular acceleration of flywheel having moment of inertia 100 kg m^2 , when a torque of 10 Nm is applied on the flywheel is:

- (1) 10 rad/s^2 (2) $\pi\text{ rad/s}^2$
(3) 0.1 rad/s^2 (4) $0.1\pi\text{ rad/s}^2$

48. A gun fires a bullet of mass 50 gm with a velocity of 30 m sec^{-1} . Because of this the gun is pushed back with a velocity of 1 m sec^{-1} . The mass of the gun is:

- (1) 15 kg (2) 30 kg
(3) 1.5 kg (4) 20 kg

49. A body of mass 5 kg explodes at rest into three fragments with masses in the ratio $1:1:3$. The fragments with equal masses fly in mutually perpendicular directions with speeds of 21 m/s . The velocity of the heaviest fragment will be:

- (1) 11.5 m/s
(2) 14.0 m/s
(3) 7.0 m/s
(4) 9.89 m/s

50. Radius of gyration of a rod of Mass M and length l is $l/2$ then distance of axis of rotation from centre of rod is:

- (1) $l/4$
(2) $l/6$
(3) $l\sqrt{6}$
(4) $l/\sqrt{6}$

SECTION-(II) CHEMISTRY**SECTION - A**

51. The modern atomic weight scale is based on:
 (1) C^{12}
 (2) O^{16}
 (3) H^1
 (4) C^{13}
52. Given below are two statements:
Statement-I: Empirical formula of methane and ethane are different.
Statement-II: Molecular formula and empirical formula of methane are different.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.
53. The volume of CO_2 produced by the combustion of 40 mL of gaseous acetone in excess of oxygen (O_2) is:
 (1) 40 mL (2) 80 mL
 (3) 60 mL (4) 120 mL
54. 0.5 mole of H_2SO_4 is mixed with 0.2 mole of $Ca(OH)_2$. The maximum number of moles of $CaSO_4$ formed is:
 $H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O$
 (1) 0.2 (2) 0.5
 (3) 0.4 (4) 1.5
55. Increasing order of electron affinity for the following configuration.
 A. $1s^2 2s^2 2p^3$
 B. $1s^2 2s^2 2p^4$
 C. $1s^2 2s^2 2p^6 3s^2 3p^3$
 D. $1s^2 2s^2 2p^6 3s^2 3p^4$
 (1) $A < C < B < D$
 (2) $D < A < C < B$
 (3) $A < B < C < D$
 (4) $A < B < D < C$
56. Pb (Atomic no. = 82) is a member of;
 (1) s-block
 (2) f-block
 (3) d-block
 (4) p-block
57. How many elements are there in 6th period of periodic table?
 (1) 18
 (2) 8
 (3) 30
 (4) 32
58. One mole of oxygen (O_2) gas at STP is equal to
 (1) 6.022×10^{23} molecules of oxygen
 (2) 6.022×10^{23} atoms of oxygen
 (3) 16 g of oxygen
 (4) 34 g of oxygen
59. What is the mole fraction of solvent in aqueous solution of NaOH having molality of 3?
 (1) 0.3 (2) 0.95
 (3) 0.7 (4) 0.05
60. Vapour pressure is the pressure exerted by vapours:
 (1) In equilibrium with liquid
 (2) In any condition
 (3) In an open system
 (4) In atmospheric conditions
61. Under which of the following conditions a gas deviates most from the ideal behaviour?
 (1) Low pressure and high temperature
 (2) High pressure and low temperature
 (3) Low temperature and low pressure
 (4) High temperature and high pressure
62. Which of the following expression at constant pressure represents Charle's law?
 (1) $V \propto \frac{1}{T}$
 (2) $V \propto \frac{1}{T^2}$
 (3) $V \propto T$
 (4) $V \propto d$

63. The IUPAC symbol for the element with atomic number 119 would be:
 (1) Unh (2) Uun
 (3) Une (4) Uue
64. On moving down the group from F to I, which of the properties decrease?
 (i) Ionic radius
 (ii) Ionisation energy
 (iii) Oxidising nature
 (iv) Electronegativity
 (1) Only (i)
 (2) (i) and (ii)
 (3) (ii), (iii), (iv)
 (4) (i), (iii), (iv)
65. The increasing order of the atomic radii of following elements is:
 (a) C (b) O
 (c) F (d) Cl
 (e) Br
 (1) (a) < (b) < (c) < (d) < (e)
 (2) (c) < (b) < (a) < (d) < (e)
 (3) (b) < (c) < (d) < (a) < (e)
 (4) (d) < (c) < (b) < (a) < (e)
66. Which of the following mixture does **not** show positive deviation from the Raoult's law?
 (1) Benzene + acetone
 (2) Acetone + ethanol
 (3) Acetone + chloroform
 (4) Water + ethanol
67. The freezing point depression constant for water is $1.86^{\circ}\text{C m}^{-1}$. If 5.00 g Na_2SO_4 is dissolved in 45.0 g H_2O , the freezing point is changed by 3.82°C . Calculate the van't Hoff factor for Na_2SO_4 .
 (1) 2.05 (2) 2.63
 (3) 3.11 (4) 0.381
68. $\text{CH}_3-\text{CH}_2-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$ and
 $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CH}_3$ are:
 (1) Functional isomers
 (2) Position isomers
 (3) Chain isomers
 (4) These are not isomers
69. Which of the following compounds is isomeric with 2, 2, 4, 4-tetramethylhexane?
 (1) 3-Ethyl-2, 2-dimethylpentane
 (2) 4-Isopropylheptane
 (3) 4-Ethyl-3-methyl-4-n-propyloctane
 (4) 4, 4-Diethyl methylheptane
70. Which of the following is ambident nucleophile?
 (1) NH_2OH
 (2) $\text{NCO}^{\ominus}$
 (3) $\text{NO}_2^{\ominus}$
 (4) All of these
71. What is the hybridisation of C-atom in CO_2 molecule?
 (1) sp^2
 (2) sp
 (3) sp^3
 (4) None of these
72. Elevation in boiling point was 0.52°C when 6 g of a non-electrolyte compound X was dissolved in 100 g of water. Molar mass of X is:
 (K_b for water = $0.52 \text{ K kg mol}^{-1}$)
 (1) 120 g mol^{-1}
 (2) 60 g mol^{-1}
 (3) 90 g mol^{-1}
 (4) 30 g mol^{-1}
73. The van't Hoff factor (*i*) **cannot** be calculated by which one of the following expressions?
 (1) $\pi V = \sqrt{i}nRT$
 (2) $\Delta T_f = iK_f \cdot m$
 (3) $\Delta T_b = iK_b \cdot m$
 (4) $\frac{P_{\text{solvent}}^{\circ} - P_{\text{solution}}}{P_{\text{solvent}}^{\circ}} = i \left(\frac{n}{N + n} \right)$
74. According to Henry's law, the solubility of a gas in a given volume of liquid increases with increase in:
 (1) Temperature
 (2) Pressure
 (3) Both (1) and (2)
 (4) None of these
75. The number of structural isomers possible from the molecular formula $\text{C}_3\text{H}_9\text{N}$ is:
 (1) 2 (2) 3
 (3) 4 (4) 5
76. The molal elevation constant is the ratio of the elevation in boiling point to:
 (1) molarity.
 (2) molality.
 (3) mole fraction of solute.
 (4) mole fraction of solvent.

77. According to Raoult's law, relative lowering of vapour pressure for a solution is equal to
- (1) Moles of solute
 - (2) Moles of solvent
 - (3) Mole fraction of solute
 - (4) Mole fraction of solvent

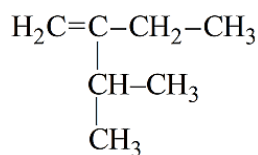
78. Which of the following will form an ideal solution?

- (1) C_2H_5OH and water
- (2) HNO_3 and water
- (3) $CHCl_3$ and CH_3COCH_3
- (4) C_6H_6 and $C_6H_5CH_3$

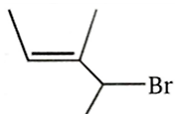
79. The correct IUPAC name from the **incorrect** name 4-Amino-3-hydroxy-2-butene is:

- (1) 1-Amino-2-hydroxy-but-2-ene
- (2) 4-Amino-but-2-en-3-ol
- (3) 1-Amino-but-2-en-2-ol
- (4) 1-Amino-but-2-en-1-ol

80. The IUPAC name of the following compound is:



- (1) 2-Ethyl-3-methylbut-1-ene
- (2) 2-Isopropylbut-1-ene
- (3) 2-Methyl-3-ethylbut-3-ene
- (4) Ethylisopropylethene

81. The IUPAC name of  is:

- (1) 2-Bromo-3-methylbut-3-ene
- (2) 4-Bromo-3-methylpent-2-ene
- (3) 2-Bromo-3-methylpent-3-ene
- (4) 4-Bromo-2, 3-dimethylbut-2-ene

82. The d-orbitals involved in sp^3d hybridisation in trigonal bipyramidal geometry:

- (1) $d_{x^2-y^2}$
- (2) d_{z^2}
- (3) d_{xy}
- (4) d_{xz}

83. Which molecule does **not** exist?

- (1) OF_2
- (2) OF_4
- (3) SF_2
- (4) SF_4

84. Which one of the following has maximum nucleophilicity in polar protic solvent?

- (1) CH_3O^-
- (2) AcO^-
- (3) $EtOH$
- (4) PhO^-

85. In one molal solution that contains 0.5 mole of a solute, there is:

- (1) 1000 g of solvent
- (2) 500 mL of solvent
- (3) 500 g of solvent
- (4) 100 mL of solvent

SECTION - B

86. In the molecular orbital diagram for the molecular ion, N_2^+ , the number of electrons in the $\sigma 2p_z$ molecular orbital is:

- (1) 0
- (2) 2
- (3) 3
- (4) 1

87. The shape of IF_7 will be

- (1) Square planar
- (2) Tetrahedral
- (3) Pentagonal bipyramidal
- (4) Irregular tetrahedral

88. Which of the following arrangements of molecules **correct** on the basis of their dipole moments?

- (1) $BF_3 > NF_3 > NH_3$
- (2) $NH_3 > BF_3 < NH_3$
- (3) $NH_3 > NF_3 > BF_3$
- (4) $NH_3 > BF_3 > NF_3$

89. How many atoms are contained in a mole of $Ca(OH)_2$?

- (1) $30 \times 6.022 \times 10^{23}$ atom/mol
- (2) $5 \times 6.022 \times 10^{23}$ atom/mol
- (3) $3 \times 6.022 \times 10^{23}$ atom/mol
- (4) $50 \times 6.022 \times 10^{23}$ atom/mol

90. An element has $[\text{Ar}]3d^1$ configuration in its +2 oxidation state. Its position in the periodic table is

- (1) period-3, group-3
- (2) period-3, group-7
- (3) period-4, group-3
- (4) period-3, group-9

91. The number of atoms present in 16 g of oxygen

(O_2) gas is:

- (1) 6.022×10^{24}
- (2) 3.01×10^{23}
- (3) 3.01×10^{24}
- (4) 6.022×10^{23}

92. The number of significant figures in 0.0045 are

- (1) Two
- (2) Three
- (3) Four
- (4) Five

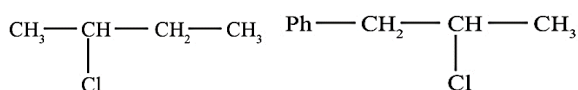
93. Which of the following species has both Covalent and coordinate bond?

- (1) H_2O
- (2) BF_4^-
- (3) NH_3
- (4) SF_2

94. Which of the following is **not** an electrophile?

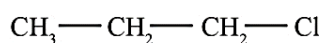
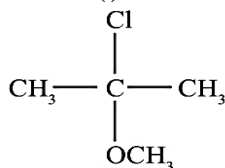
- (1) CH_3^+
- (2) AlCl_3
- (3) H_3O^+
- (4) $\left[(\text{CH}_3)_4\text{N}^+\right]^\ominus\text{OH}$

95. Reactivity order for $\text{S}_\text{N}1$ will be:



(i)

(ii)



(iii)

(iv)

- (1) (i) > (ii) > (iii) > (iv)
- (2) (ii) > (i) > (iii) > (iv)
- (3) (iii) > (ii) > (i) > (iv)
- (4) (iv) > (iii) > (ii) > (i)

96. Match List-I with List-II.

List-I		List-II	
(A)	XeF_6	(I)	Distorted octahedral
(B)	XeO_3	(II)	Square planar
(C)	XeOF_4	(III)	Pyramidal
(D)	XeF_4	(IV)	Square pyramidal

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-IV, B-I, C-II, D-III

97. The **incorrect** statement among the following.

- (1) The first ionisation enthalpy of Al is less than the first ionisation enthalpy of Mg
- (2) The second ionisation enthalpy of Mg is greater than the second ionisation enthalpy of Na
- (3) The first ionisation enthalpy of Na is less than the first ionisation enthalpy of Mg
- (4) The third ionisation enthalpy of Mg is greater than third ionisation enthalpy of Na

98. For a fixed volume of a gas the pressure is increases when the temperature increases. This is due to:

- (1) Increase in the average molecular speed
- (2) Increase in the rate of collisions
- (3) Increase in the molecular attraction
- (4) Decrease in the mean free path

99. Which of the following is Raoult's law **not** applicable?

- (1) 1 m NaCl
- (2) 1 m glucose
- (3) 1 m Urea
- (4) 1 m sucrose

100. Match List-I with List-II.

List-I		List-II	
(A)	$\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O}$	(I)	Molal elevation constant
(B)	Ebullioscopic constant	(II)	Molal depression constant
(C)	Cryoscopic constant	(III)	Non-ideal solution with positive deviation
(D)	$\text{H}_2\text{O} + \text{H}_2\text{SO}_4$	(IV)	Non-ideal solution with negative deviation

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-I, D-III
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-II, C-I, D-IV
- (4) A-IV, B-I, C-II, D-III

SECTION-(III) BOTANY**SECTION - A****101. Match List-I with List-II:**

List-I		List-II	
(A)	Saprophyte	(I)	Symbiotic association of fungi with plant roots
(B)	Parasite	(II)	Decomposition of dead organic materials
(C)	Lichens	(III)	Depend on living plants or animals
(D)	Mycorrhiza	(IV)	Symbiotic association of algae and fungi

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-II, B-I, C-III, D-IV

102. Which of the following is against the rules of ICBN?

- (1) Generic and specific epithet should start with small letters.
- (2) Hand written scientific names should be underlined.
- (3) Every species should have a generic name and a specific epithet.
- (4) Scientific names are in Latin and should be written in italic.

103. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Sexual reproduction in *Volvox* is oogamous type.

Reason R: Both gamete of *Volvox* are non-flagellated and similar in size.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

104. Select the **incorrect** from the following.

- (1) *Petunia* – Monocotyledonae
- (2) *Triticum* – Poales
- (3) *Datura* – Polymoniales
- (4) *Solanum* – Solanaceae

105. Which of the given statements is **not** true for this fungus?

- (1) It is commonly known as a sac-fungus.
- (2) It produces conidia endogenously.
- (3) Its mycelium is branched and septate.
- (4) It forms fruiting body called ascocarp.

106. Carnivora includes:

- (1) group of organisms belonging to related genera.
- (2) group of organisms belonging to related species.
- (3) group of organisms belonging to related families.
- (4) group of organisms which are similar in all features.

107. Which of the following statements is **incorrect**?

- (1) Morels and truffles are edible delicacies.
- (2) *Claviceps* reproduce through conidia.
- (3) Ascospores are produced endogenously.
- (4) Yeasts have filamentous bodies with long thread-like hyphae.

108. Read the following and choose the **correct** statement.

- (1) The thalloid plant body of liverwort is dorsiventrally appressed closely to substrate.
- (2) The leafy members have tiny true leaf in two rows on the stem like structures.
- (3) The leafy member has tiny leaf like appendage in four rows on the stem like structures.
- (4) The thalloid plant body of liverwort is isobilaterally appressed closely to substrate.

109. Read the following statements and select the option accordingly.

- A. In prokaryotes, ribosomes are associated with the plasma membrane of the cell.
- B. Several ribosomes on a single mRNA form a structure called polysome.
- C. Polysome translates the mRNA into proteins.
- D. Gas vacuoles are found in blue green and purple and green photosynthetic bacteria.

Choose the most appropriate answer from the options given below:

- (1) Only D is incorrect
- (2) Only A and B are incorrect
- (3) A, B and C are correct only
- (4) A, B, C and D are correct

- 110.** Choose the **incorrect** statement.
- (1) Matthias Schleiden examined a large number of plants.
 - (2) Schwann studied different types of animal cells.
 - (3) Schleiden observed that all plants are composed of the same types of cells which form the tissues of the plant.
 - (4) Theodore Schwann was a British Zoologist and Matthias Schleiden was a German botanist.

- 111.** As we go from kingdom to species, the number of common characteristics:
- (1) remains constant.
 - (2) decrease.
 - (3) increase.
 - (4) show fixed patterns of increase and decrease.

- 112.** In the list given below maximum members are of which kingdom?

Nostoc, Anabaena, Amoeba, Entamoeba, Ustilago, Claviceps and Colletotrichum

Choose the most appropriate answer from the options given below:

- (1) Protista
- (2) Monera
- (3) Fungi
- (4) Plantae

- 113.** Given below are two statements:

Statement I: Name of the author in the biological name is written in an abbreviated form.

Statement II: 'Animals', 'mammals' and 'dogs' represent taxa at same levels.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

- 114.** How many of the following is/are **not** considered as part of the endomembrane system?

- A. Golgi complex
- B. Peroxisome
- C. Vacuole
- D. Lysosome

Choose the most appropriate answer from the options given below:

- (1) A and C only
- (2) A and D only
- (3) A and B only
- (4) B only

- 115.** Identify the meiotic stage in which the homologous chromosomes separate while the sister chromatids remain associated at their centromeres.

- (1) Anaphase-II
- (2) Metaphase-I
- (3) Prophase-I
- (4) Anaphase-I

- 116.** Match **List-I** with **List-II**.

List-I		List-II	
(A)	Metacentric chromosome	(I)	Centromere is situated close to chromosome end forming one extremely short and one very long arm
(B)	Sub-metacentric chromosome	(II)	Middle centromere forming two equal arms of the chromosome
(C)	Acrocentric chromosome	(III)	Centromere slightly away from the middle of the chromosome
(D)	Telocentric chromosome	(IV)	Terminal centromere on the chromosome

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-I, D-IV

- 117.** Select the **wrong** statement.

- (1) Pseudopodia are locomotory and feeding structures in *Amoeboid protozoan*.
- (2) Mushrooms belong to basidiomycetes.
- (3) Cell wall is present in members of fungi and plantae.
- (4) Mesosome is present in all kingdoms except monera.

- 118.** Given below are two statements:

Statement I: Phosphate granules are example of inclusion bodies.

Statement II: Cyanophycean granules and glycogen granules are not bound by any membrane.

In the light of the above statements, choose the most appropriate answer from the option given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

119. Choose the **incorrect** statement from the following.

- (1) Unicellular organisms are capable of independent existence.
- (2) Anything less than a complete structure of a cell does not ensure independent living.
- (3) Cell is the fundamental structural and functional unit of all non-living organisms.
- (4) All cells arise from pre-existing cells.

120. Brinjal belongs to the genus:

- (1) *Mangifera*. (2) *Solanum*.
- (3) *Triticum*. (4) *Petunia*.

121. Average duration of cell cycle of a human cell is:

- (1) 12 hour. (2) 16 hour.
- (3) 20 hour. (4) 24 hour.

122. Read the following four statements (A-D) carefully.

- A. Mycoplasma can not survive without oxygen
- B. Mycoplasma are the smallest living cells known.
- C. Bacteria completely lack a cell wall.
- D. Bacteria reproduce mainly by fission.

How many of the above statements are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

123. Gorilla belongs to all **except**:

- (1) order-Primata.
- (2) phylum-Chordata.
- (3) kingdom-Animalia.
- (4) class-Anacardiaceae.

124. Select an **incorrect** statement about gemmae in bryophytes?

- (1) Gemmae are green, multicellular, sexual buds.
- (2) Gemmae develop in small receptacles called gemma cups located on the thalli.
- (3) Gemmae become detached from the parent body and germinate to form new individuals.
- (4) Gemmae are green, multicellular, asexual buds.

125. Oogamous and non-motile gamete with chlorophyll *a* present in algae:

- (1) *Laminaria*.
- (2) *Chlorella*.
- (3) *Porphyra*.
- (4) *Spirogyra*.

126. In which stage of mitosis chromosomes are arranged on the equator of spindle?

- (1) Interphase (2) Anaphase
- (3) Prophase (4) Metaphase

127. Select the **wrong** statement.

- (1) Diatoms are chief producers in the oceans.
- (2) Diatoms are microscopic and float passively in water.
- (3) The walls of diatoms are easily destructible.
- (4) 'Diatomaceous earth' is formed by the cell walls of diatoms.

128. Mature heart cells cannot undergo cell division. These cells are probably considered in:

- (1) G₂ phase.
- (2) S phase.
- (3) mitosis.
- (4) G₀ phase.

129. Select the **correct** statement from the following.

- A. Plants show continuous growth throughout their life.
- B. Mitosis occurs in apical meristematic tissues.
- C. A very significant contribution of mitosis is cell repair.
- D. Lateral cambium divides by mitosis.

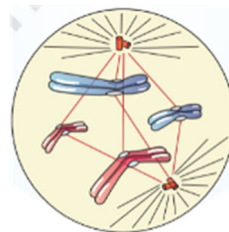
Choose the most appropriate answer from the options given below:

- (1) C only
- (2) B and C only
- (3) A and C only
- (4) A, B, C and D

130. ICZN stands for:

- (1) Indian Code of Zoological Names.
- (2) Indian Code of Zoological Nomenclature.
- (3) International Code of Zoological Names.
- (4) International Code of Zoological Nomenclature.

131. Given below is the representation of a certain event at a particular stage of mitosis. Which stage will occur after this stage?



- (1) Prophase of mitosis.
- (2) Telophase.
- (3) Metaphase.
- (4) G₀ phase.

132. Consider the following taxa (A to D) and find out the **correct** sequence of taxonomical categories from lower to higher category.
- Petunia*
 - Polymoniales
 - Dicotyledonae
 - Solanaceae
- A→D→B→C
 - C→D→B→A
 - A→B→C→D
 - C→B→A→D
133. Which stage in mosses bears the sex organs?
- Secondary protonema
 - Protonema stage
 - Leafy stage
 - Sporophyte
134. In bryophytes, male sex organ is:
- unicellular antheridium, biflagellate.
 - multicellular antheridium, biflagellate.
 - multicellular antheridium, multiflagellate.
 - unicellular spermatogonium, non-motile.
135. In the taxonomic categories of man and housefly, what is the equivalent rank of Poales for each respectively?
- Homo* and *Musca*.
 - Hominidae and Muscidae.
 - Mammalia and Insecta.
 - Primata and Diptera.

SECTION - B

136. Interphase includes all of the following **except**:
- G₁ phase.
 - anaphase.
 - S phase.
 - G₂ phase.

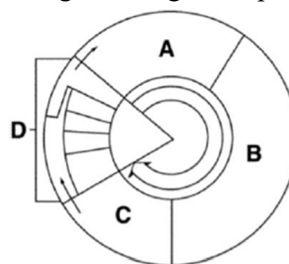
137. Match List-I with List-II:

List-I		List-II	
(A)	Slime layer	(I)	Strong structural support
(B)	Capsule	(II)	Thick and tough
(C)	Cell wall	(III)	Loose sheath
(D)	Cell membrane	(IV)	Interacts with the outside world

Choose the **correct** answer from the options given below:

- A-III, B-II, C-I, D-IV
- A-II, B-III, C-IV, D-I
- A-III, B-IV, C-I, D-II
- A-II, B-III, C-I, D-IV

138. Sporophyte in bryophytes is:
- free-living.
 - attached to the photosynthetic gametophyte.
 - unicellular.
 - produced by spores.
139. In which of the following, gametophyte is **not** independent free living?
- Pteris*
 - Pinus*
 - Funaria*
 - Marchantia*
140. Taxonomy takes into account:
- identification only.
 - nomenclature and identification of organisms only.
 - evolutionary relationship only.
 - characterisation, identification, nomenclature and classification.
141. The given diagram represents stages of cell cycle.



Identify which of the following is **correct** labelling for A, B, C or D marked in the given figure.

- A-G₁ phase
- B-G₂ phase
- C-G₃ phase
- D-G₀ phase

142. Select the **correct** statements for endoplasmic reticulum.

- Rough endoplasmic reticulum is frequently observed in the cells actively involved in protein synthesis.
- Rough endoplasmic reticulum is frequently observed in the cells actively involved in secretion.
- Smooth endoplasmic reticulum are extensive and continuous with the outer membrane of the vacuoles.
- In animal cells lipid-like steroidal hormones are synthesised in Smooth endoplasmic reticulum.

Choose the most appropriate answer from the options given below:

- A only
- A and B only
- A, B and D only
- A and C only

143. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The scientific name of lion is written as *Panthera leo*.

Reason R: Lion belongs to Felidae family.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

144. The small bristle like fibres sprouting out of the bacterial cell are:

- (1) pili.
- (2) filament.
- (3) hook.
- (4) fimbriae.

145. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Canidae	(I)	Phylum
(B)	Sapindales	(II)	Order
(C)	Insecta	(III)	Class
(D)	Arthropoda	(IV)	Family

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-IV, B-II, C-III, D-I
- (3) A-III, B-II, C-I, D-IV
- (4) A-I, B-II, C-IV, D-III

146. Identify the meiotic stage in **A**, **B** and **C** options given below.

- A. The chromosome becomes gradually visible with compaction of chromatin.
- B. The complex formed by a pair of synapsed homologous chromosomes.
- C. Recombination nodules are seen.

Choose the most appropriate answer from the options given below:

	A	B	C
(1)	Leptotene	Pachytene	Diplotene
(2)	Leptotene	Zygotene	Pachytene
(3)	Zygotene	Pachytene	Diplotene
(4)	Zygotene	Diplotene	Pachytene

147. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion: The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins.

Reason: Chloroplast contains small, double stranded circular DNA molecules and ribosomes.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

148. Chemotaxonomy is connected with:

- (1) use of the chemical constituents of the plant.
- (2) use of phylogenetic relationship between organisms.
- (3) observable characteristics.
- (4) chromosome behaviour.

149. Read the following and identify the **incorrect** statements.

- A. The lipids are arranged within the cell membrane with the polar tail towards the outer sides and the hydrophobic head towards the inner part.
- B. In cell membranes, the peripheral proteins lie on the surface of the membrane while the integral proteins are partially or totally buried in the membrane.
- C. According to the fluid mosaic model, the quasi-fluid nature of lipids enables lateral movement of proteins within the overall bilayer.
- D. Fluid mosaic model of the structure of cell membrane was proposed by Singer and Nicolson (1982).

Choose the most appropriate answer from the options given below:

- (1) A and D only
- (2) A only
- (3) B only
- (4) B and C only

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Classification is the process by which anything is grouped into convenient categories based on some easily observable characters.

Reason R: It is nearly impossible to study all the living organisms.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

SECTION-(IV) ZOOLOGY**SECTION - A**

151. Which of the following options is **incorrect** w.r.t double stranded helical structure of DNA?

- (1) Adenine pairs with thymine.
- (2) Cytosine pairs with guanine.
- (3) Uracil pairs with adenine.
- (4) Deoxyribose sugar is present.

152. Mucus, digestive enzymes and milk are secreted by;

- (1) exocrine glands.
- (2) endocrine glands.
- (3) composite glands.
- (4) all of these

153. Bones are connected to another bone by means of;

- (1) cartilage.
- (2) ligament.
- (3) tendon.
- (4) adipose tissue.

154. Which of the following is a carbohydrate in nature?

- (1) Receptor
- (2) Starch
- (3) Trypsin
- (4) Antibody

155. Match **List-I** with **List-II** w.r.t cockroach.

List-I		List-II	
(A)	Malpighian tubules	(I)	Used in flight
(B)	Hepatic caeca	(II)	100 – 150
(C)	Forewings	(III)	Opaque dark
(D)	Hindwings	(IV)	6 – 8

Choose the **correct** answer from the options given below:

- (1) A-I, B-II C-III, D-IV
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-I, C-III, D-IV
- (4) A-I, B-II, C-IV, D-III

156. Statocysts are:

- (1) balancing organs of arthropods.
- (2) respiratory organs of arthropods.
- (3) excretory organs of arthropods.
- (4) sex organs of annelids.

157. Given below are two statements.

Statement I: Endoskeleton of cockroach has hardened plates called sclerites.

Statement II: Sclerites are joined together by arthrodial membrane.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

158. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Blood	(I)	Outer ear joints
(B)	Lacunae	(II)	Areolar tissue
(C)	Cartilage	(III)	Osteocytes
(D)	Fibroblast	(IV)	White blood cells

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-I, C-II, D-IV
- (4) A-II, B-III, C-I, D-IV

159. Excess nutrients which are **not** used immediately are converted into fats and stored in;

- (1) muscle tissue.
- (2) adipose tissue.
- (3) blood.
- (4) bone.

160. Identify the types of cell junctions that perform cementing to keep neighbouring cells together.

- (1) Tight junctions
- (2) Adhering junctions
- (3) Gap junctions
- (4) All of these

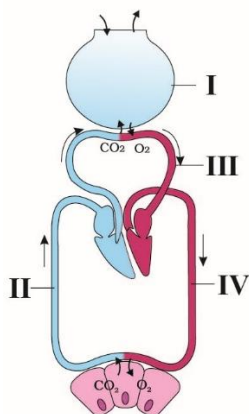
161. Michaelis constant (K_m) value of an enzyme is the substrate concentration at which the velocity of reaction reaches:

- (1) V_{max} .
- (2) $1/3 V_{max}$.
- (3) $1/2 V_{max}$.
- (4) $1/5 V_{max}$.

162. Select the **incorrect** statement w.r.t frog.
- (1) Fertilisation is internal and takes place inside the body.
 - (2) A mature female can lay 2500 to 3000 ova at a time.
 - (3) Frogs have renal portal system.
 - (4) Frogs exhibit mimicry.

163. Select the **correct** statement w.r.t neurons.
- (1) They exert greatest control over body's responsiveness to changing condition.
 - (2) They are excitable cells.
 - (3) It is the fluid connective tissue.
 - (4) Both (1) and (2)

164. Refer to the following diagram and identify the blood vessel that carries deoxygenated blood from body tissue and having pO_2 is 40 mmHg.



- (1) IV
- (2) I
- (3) II
- (4) III

165. Phosphodiester bond is present between:
- (1) amino acids in a polypeptide chain.
 - (2) nucleotides in a nucleic acid.
 - (3) fatty acids in a triglyceride.
 - (4) monosaccharides in a polysaccharide.

166. Choose the **incorrect** statement.
- (1) Long exposure to dust can cause emphysema.
 - (2) The nasal chamber opens into pharynx.
 - (3) Alveoli are very thin, irregular-walled and vascularised bag-like structures.
 - (4) Solubility of gases affects rate of diffusion.

167. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Fishes	(I)	Through lungs
(B)	Insects	(II)	Through skin
(C)	Human	(III)	Through a network of tracheal tubes
(D)	Amphibians	(IV)	Through gills

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-I, C-III, D-II
- (3) A-II, B-IV, C-I, D-III
- (4) A-IV, B-III, C-II, D-I

168. Read the following statements (A-D).

- A. The inner pleural membrane is in close contact with the thoracic lining.
- B. The outer pleural membrane is in contact with the lung surface.
- C. The part starting with the external nostrils up to the terminal bronchioles constitute the exchange part.
- D. The maximum volume of air a person can breathe in after a forced expiration is the total lung capacity.

Which of the above statements is/are **incorrect**?

- (1) A, B and C only
- (2) B and D only
- (3) A, B and D only
- (4) A, B, C and D

169. The sinus venosus in frogs:

- (1) joins the left atrium.
- (2) joins the right atrium.
- (3) joins the ventricle directly.
- (4) joins the conus arteriosus.

170. Which of the following compound is **not** found in the acid-insoluble pool?

- (1) Proteins
- (2) Lipid
- (3) Nucleic acid
- (4) Glucose

171. Trachea is a straight tube extending up to the mid-thoracic cavity which divides at the level of ____.

Choose the option which fills the blank **correctly**.

- (1) 5th thoracic vertebrae.
- (2) 4th thoracic vertebrae.
- (3) 3rd thoracic vertebrae.
- (4) 7th thoracic vertebrae.

172. The volume of O_2 delivered by 100 mL of oxygenated blood to the tissues is;

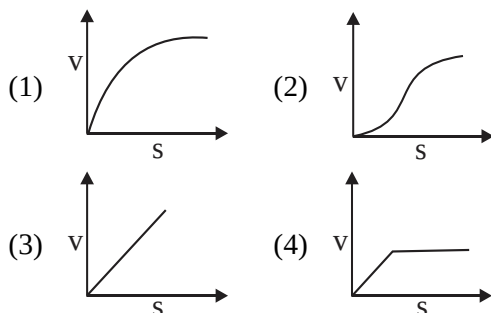
- (1) 4 mL.
- (2) 5 mL.
- (3) 15 mL.
- (4) 20 mL.

173. Neural signal from which of the following can reduce the duration of inspiration?

- (1) Chemosensitive area
- (2) Cardiovascular rhythm centre
- (3) Pneumotaxic centre
- (4) Receptors associated with aortic arch and carotid artery

174. Contraction of external intercostal muscles increases the volume of thoracic chamber in the;
- (1) antero-posterior axis.
 - (2) dorso-ventral axis.
 - (3) ventro-lateral axis.
 - (4) dorso-lateral axis.

175. Which of the following graphs best explain the relationship between rate of an enzymatic reaction and substrate concentration?



176. Which of the following phylum has a metamerically segmented body?
- (1) Chordata
 - (2) Arthropoda
 - (3) Annelida
 - (4) All of these

177. Given below are two statements:

Statement I: Hemichordata consists of worm like marine animals.

Statement II: The body of Hemichordates is divided into three regions: proboscis, collar, and trunk.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is incorrect but Statement II is correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I and Statement II both are correct.
- (4) Statement I and Statement II both are incorrect.

178. Which of the following group comprises exclusively marine animals?
- (1) Annelida and Mollusca
 - (2) Arthropoda and Echinodermata
 - (3) Echinodermata and Hemichordata
 - (4) Ctenophora and Coelenteratea

179. All of the following animals belong to phylum Mollusca **except**:
- (1) *Aplysia*.
 - (2) *Cucumaria*.
 - (3) *Loligo*.
 - (4) *Sepia*.

180. Identify the **correct** option for Aschelminthes.
- (1) Triploblastic, coelomate with organ system level of organisation.
 - (2) Diploblastic, acoelomate with tissue level of organisation.
 - (3) Triploblastic, acoelomate with organ system level of organisation.
 - (4) Triploblastic, pseudocoelomate with organ system level of organisation.

181. Which of the following is an **incorrect** match?

- (1) *Euspongia* - Asymmetrical
- (2) *Adamsia* - Diploblastic
- (3) *Pleurobrachia* - Tissue level of organisation
- (4) *Nereis* - Pseudocoelomate

182. Read the following statements (I-V).

- I. *Pteropus* is an oviparous mammal.
- II. Ostrich is a flightless bird.
- III. Incomplete digestive system is present in Cnidaria.
- IV. Stomochord is present in the *Saccoglossus*.
- V. *Echinus* is a fresh water sponge.

Which of the above statements are **incorrect**?

- (1) I and II only
- (2) I and V only
- (3) II and III only
- (4) IV and V only

183. Which of the following statements about arthropods is **incorrect**?

- (1) It is the largest phylum of Kingdom Animalia.
- (2) Arthropods include insects, spiders, and crustaceans.
- (3) Jointed appendages are present.
- (4) Excretion takes place by flame cells.

184. Identify the phylum on the basis of characteristics given.

- A. Balancing organs are present.
 - B. Body is covered by a chitinous exoskeleton.
 - C. Body consists of head, thorax and abdomen.
 - D. Circulatory system is of open type.
- (1) Aschelminthes
 - (2) Porifera
 - (3) Arthropoda
 - (4) Hemichordata

185. *Aurelia* belongs to phylum:

- (1) Porifera.
- (2) Coelenterata.
- (3) Platyhelminthes.
- (4) Ctenophora.

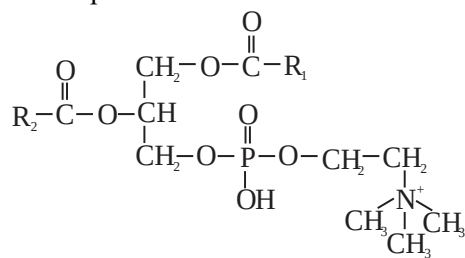
SECTION - B

186. How many of the following organisms show a closed type of blood circulation?

Laccifer, Nereis, Limulus, Pheretima, Aedes
mosquito, *Saccoglossus, Naja*

- (1) Six
(2) Five
(3) Four
(4) Three
187. Which of the following statements is **correct**?
- (1) Female cockroach possesses fifteen ovarioles in the ovaries.
(2) Cockroaches exhibit mosaic vision with less sensitivity and more resolution.
(3) A mushroom-shaped gland is present in the 7th – 8th abdominal segments of male cockroach.
(4) A pair of spermatheca is present in the 6th segment of female cockroach.
188. In which of the following class of organism external fertilisation occurs?
- (1) Amphibia
(2) Chondrichthyes
(3) Mammalia
(4) Aves
189. All of the following statements are correct, **except**:
- (1) Starch produce color with I₂.
(2) N end of polypeptide chain is on the left hand side.
(3) Right end is non-reducing in a polysaccharide chain.
(4) Cellulose consists of only one type of monosaccharide.
190. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Frogs have the ability to change the colour.
Reason R: This protective coloration is called poikilotherms.
In the light of the above statements, choose the correct answer from the options given below.
- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

191. Identify the below given molecule and choose the **correct** option.



- (1) A nucleotide
(2) A diglyceride
(3) A nucleoside
(4) A phospholipid
192. Read the following statement (A-E).
- A. Abrin is a secondary metabolite.
B. GLUT-4 enables glucose transport into cells.
C. Glycogen is a polymer of galactose.
D. Ricin is a toxin.
E. The complex polysaccharides are mostly heteropolymers.
- Which of the above statements are **correct**?
- (1) A, B and D only
(2) A, C and D only
(3) B and D only
(4) A, B, C and E only
193. $\text{CO}_2 + \text{H}_2\text{O} \xrightleftharpoons{\text{X}} \text{H}_2\text{CO}_3 \xrightleftharpoons{\text{Y}} \text{HCO}_3^- + \text{H}^+$
Choose the **correct** answer from the options given below:

	X	Y
(1)	Carbonic anhydrase	Carbonic anhydrase
(2)	Carbonic anhydrase	Carboxypeptidase
(3)	Carboxy peptidase	Carbonic anhydrase
(4)	Carbonic anhydrase	Transferase

194. Complete the analogue and choose the **correct** option.
Anthocyanin: Pigment :: Vinblastin : _____
- (1) Drug
(2) Essential oil
(3) Lectin
(4) Alkaloid
195. How many of the following structures are supported by incomplete cartilaginous rings?

Secondary bronchi, Tertiary bronchi, Trachea, Pharynx, initial bronchioles

- (1) Three
(2) Four
(3) Five
(4) Two

196. Match **List-I** with **List-II**:

	List-I		List-II
(A)	Apoenzyme	(I)	Quaternary structure of protein
(B)	NADP	(II)	Malonate
(C)	Inhibitor of succinate dehydrogenase	(III)	Protein part of the enzyme.
(D)	Haemoglobin	(IV)	Niacin

Choose the **correct** answer from the options given below.

- (1) A-I, B-II, C-III, D-IV
- (2) A-IV, B-II, C-I, D-III
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-I, C-II, D-IV

197. Which of the following pair have similar kind of epithelial tissue?

- (1) Wall of blood vessels and alveoli
- (2) Lining of stomach and intestine
- (3) Moist surface of buccal cavity and pharynx
- (4) All of these

198. Given below are two statements.

Statement I: Muscular tissues are most abundant in the body.

Statement II: Blood is an example of specialised connective tissue.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

199. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Rate of the chemical reaction triples on increasing the temperature by 10° celsius.

Reason R: Catalysed reactions proceed at rates higher than that of uncatalysed reactions.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

200. Tidal volume and expiratory reserve volume of an athlete is 700 mL and 1000 mL respectively. What will be his expiratory capacity if the residual volume is 1200 mL?

- (1) 1500 mL
- (2) 1700 mL
- (3) 2200 mL
- (4) 2900 mL

